SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***SK ALI BABU(192524038)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:SK.ALIBABU

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.

The **Median Filter** is the most suitable filtering technique for removing salt-and-pepper noise. It effectively removes impulse noise by replacing each pixel with the median value of its neighboring pixels while preserving important edge information.

b. Explain why the selected filter performs better than an averaging filter for impulse noise removal.

The averaging (mean) filter replaces the pixel value with the average of its neighboring pixels. Since salt-and-pepper noise consists of extreme intensity values (0 or 255), averaging spreads the noise to neighboring pixels and causes image blurring. In contrast, the median filter replaces the noisy pixel with the median value of the surrounding pixels, thereby removing impulse noise without significantly affecting image edges.

c. If a 3 × 3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.

- Noise density = 0.05
- Noise is randomly distributed.

- Each pixel is independently corrupted.
- A noisy center pixel is corrected if at least five pixels in the 3×3 neighborhood are noise-free.

The probability is given by:

$P(\text{Corrected}) = \Sigma(k = 0 \text{ to } 4)\ C(8, k) \times (0.05)^k \times (0.95)^{(8 - k)}$

$P(\text{Corrected}) \approx 0.9959$

(Approximately 99.59%)

d. Estimate the total number of noisy pixels present in the original image.

Given:

Image size = 1024×1024 pixels

Noise density = 0.05 (5%)

Calculation:

Total number of pixels

= 1024×1024

= 1,048,576 pixels

Total noisy pixels

= Total Pixels × Noise Density

= $1,048,576 \times 0.05$

= $52,428.8 \approx 52,429$ pixels

Therefore, the original image contains approximately 52,429 noisy pixels.

e. Estimate the expected number of noisy pixels remaining after median filtering.

Given:

Total noisy pixels = 52,429

Probability of correction = 99.59% = 0.9959

Calculation:

Probability that a noisy pixel remains uncorrected

= 1 – 0.9959

= 0.0041

Expected remaining noisy pixels

= 52,429 × 0.0041

= 214.96 ≈ 215 pixels

Therefore, approximately 215 noisy pixels are expected to remain after median filtering.

f. Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

If the image is affected by Gaussian noise, a **Gaussian Filter** or **Bilateral Filter** should be used instead of a median filter. Gaussian filters effectively reduce Gaussian noise while preserving image quality. Bilateral filters further preserve edges during smoothing. Median filters are specifically designed for impulse noise and are less effective for Gaussian noise.

Question 2: Discretization of a Continuous Image

a. Explain how the continuous image can be converted into a digital image using sampling and quantization.

A continuous image is converted into a digital image through two processes:

- **Sampling:** Converts continuous spatial coordinates into discrete pixel locations.
- **Quantization:** Converts continuous intensity values into a finite number of gray levels.

b. Identify the spatial variation pattern of the image in the x-direction.

The image exhibits a **sinusoidal intensity variation** along the x-direction, while the intensity remains constant along the y-direction.

c. Determine the minimum sampling requirement needed to avoid aliasing in the x-direction

According to the Nyquist Sampling Theorem, the sampling frequency must be at least twice the highest spatial frequency present in the image.

Formula:

$f_s \geq 2f_x$

where,

f_s = Sampling frequency

f_x = Highest spatial frequency of the sinusoidal signal

Minimum sampling requirement: $f_s \geq 2f_x$.

d. Explain what will happen if the image is sampled below the required sampling rate.

Sampling below the Nyquist rate results in **aliasing**, where high-frequency components appear as lower frequencies. This produces distorted patterns, loss of image detail, and inaccurate image representation.

e. Suggest a suitable 8-bit quantization scheme for storing the image.

A **uniform 8-bit quantization** scheme is suitable.

- Number of gray levels = 256
- Intensity range = 0–255

This provides sufficient intensity resolution for grayscale image storage.

f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Higher quantization levels improve image quality and intensity accuracy but require more storage space. Lower quantization levels reduce storage requirements but introduce quantization errors and visible intensity banding.

Question 3: Edge Detection in Thin Line Images

a. Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.

Edge Detector	Advantages	Disadvantages	Suitability
Sobel	Good noise suppression and edge detection	Slight edge blurring	Good
Prewitt	Simple and computationally efficient	Sensitive to noise	Moderate
Canny	Excellent localization and noise suppression	Higher computational cost	Best

b. For a horizontal edge located near pixel position (256,256), compute the Sobel gradient response.

Given:

$Gx = 0$

$Gy = 1020$

Gradient Magnitude:

$|G| = \sqrt{(Gx^2 + Gy^2)}$

$= \sqrt{(0^2 + 1020^2)}$

$= \sqrt{1,040,400}$

$= 1020$

Gradient Direction:

$\theta = \tan^{-1}(Gy / Gx)$

$= \tan^{-1}(1020 / 0)$

$= 90°$

Gradient Magnitude = 1020

Gradient Direction = 90°

c. Explain how the gradient magnitude and gradient direction help identify an edge.

The **gradient magnitude** represents the strength of the edge, whereas the **gradient direction** indicates the orientation of the edge within the image.

d. Discuss the role of Gaussian smoothing in the Canny edge detection process.

Gaussian smoothing removes image noise before gradient computation, thereby reducing false edge detection and improving edge localization.

e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.

A small Gaussian kernel preserves thin edges, whereas a large Gaussian kernel blurs thin lines and reduces edge localization accuracy.

f. Explain the effect of reducing the Canny threshold by half.

Reducing the threshold increases the number of detected edges, allowing weak edges to be identified. However, it also increases the detection of false edges caused by noise.

Question 4: Feature Extraction for Handwritten Digit Recognition

a. Explain the working principles of SIFT, SURF, and ORB.

- **SIFT (Scale Invariant Feature Transform):** Detects scale-space keypoints using Difference of Gaussian and generates robust feature descriptors.
- **SURF (Speeded-Up Robust Features):** Uses Hessian matrix approximation and integral images for faster feature detection.
- **ORB (Oriented FAST and Rotated BRIEF):** Combines FAST keypoint detection with BRIEF descriptors to provide fast and efficient feature extraction.

b. Compare SIFT, SURF, and ORB.

Property	SIFT	SURF	ORB
Scale Invariance	Yes	Yes	Yes
Rotation Invariance	Yes	Yes	Yes
Computational Cost	High	Medium	Low
Real-Time Performance	No	Moderate	Yes

c. If a digit image is rotated by 45°, identify which methods still produce reliable features.

All three methods—**SIFT, SURF, and ORB**—are rotation invariant and can produce reliable matching features for a 45° rotated image.

d. Compute the total number of keypoints detected.

Average keypoints per image = 120

Number of images = 50

120 * 50 = 6000

Therefore, the total number of detected keypoints is **6000**.

e. Suggest a suitable dimensionality reduction technique.

Principal Component Analysis (PCA) is an appropriate dimensionality reduction technique that retains approximately **95%** of the useful information while reducing feature dimensions.

f. Explain how dimensionality reduction improves classification efficiency.

Dimensionality reduction decreases computational complexity, reduces memory usage, removes redundant information, and improves classification speed and accuracy.

Question 5: Morphological Preprocessing for Character Recognition

a. Identify the morphological operation suitable for removing small white noise dots.

Morphological Opening (Erosion followed by Dilation) is used to remove small white noise from the background.

b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.

Morphological Closing (Dilation followed by Erosion) is used to fill small gaps and holes inside character strokes.

c. Explain the difference between opening and closing operations.

Opening removes small foreground noise and smooths object boundaries, whereas Closing fills small holes and connects broken parts of the object.

d. Explain the effect of using a 3 × 3 square structuring element.

A 3 × 3 structuring element removes isolated noise pixels while preserving the general shape of the characters. However, extremely thin strokes may become slightly thinner.

e. Discuss the risk of using a large structuring element.

A large structuring element may remove important character details, shrink object boundaries, merge nearby characters, and distort thin strokes, thereby reducing recognition accuracy.

f. Propose a complete preprocessing pipeline for handwritten character recognition.

1. Image Acquisition (Scanner/Camera)
2. Grayscale Conversion
3. Noise Removal
4. Image Binarization (Thresholding)
5. Morphological Opening
6. Morphological Closing
7. Size Normalization
8. Feature Extraction (SIFT, SURF, ORB, HOG, etc.)
9. Character Classification (SVM, KNN, CNN)